



**FAST National University of Computer and
Emerging Sciences**

Report: Thesis I

***Towards Improving Deep Embedded Document
Clustering with long range text semantics and
neighborhood documents information***

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1. Introduction & Background

Document clustering is an unsupervised machine learning problem, which aims to learn latent semantic patterns from the text data and organize unlabeled documents into semantically coherent clusters. Deep learning-based document clustering algorithms recently performed very well. These algorithms try to learn the features and clustering arrangement simultaneously and play an important role in information retrieval, topic extraction and recommendation systems and large-scale content management. However, it is difficult to cluster textual data due to the high dimensionality of document representations and the difficulty to capture meaningful semantic relationships among documents.

Recent research in deep embedded clustering (DEC) frameworks have improved clustering performance by learning feature representations and clustering assignments simultaneously. Models such as SDCN [1], DSEDC [2], and AGCN [3] proved that integrating structural relationships with graph convolutional networks (GCNs) improves clustering quality. Likewise, transformer-based approaches such as SDEC [4] and DTSN [5] enhanced semantic representation by capturing long-range contextual dependencies between documents. Despite these advancements, there are two major challenges.

First, many graph-based clustering models heavily rely on fixed graph structure constructed early in training. These static structural assumptions may introduce early-stage noise and fail to capture as semantic embeddings evolve during optimization. Second, although semantic and structural representations are often optimized simultaneously, explicit mechanisms to enforce consistency between these two views are rare.

These limitations reveal a fundamental bottleneck in deep embedded document clustering, namely maintaining an effective balance between semantic representation learning and structurally guided clustering alignment while ensuring robustness.

2. Objectives

To address the aforementioned limitations, the thesis proposes two major objectives:

- ❖ Obtain the long-range semantic representations using a Transformer guided semantic representation refinement, thus improving the representation of the content of documents.
- ❖ Use neighborhood structural information via adaptive graphs construction and Graph Convolutional Network (GCN) modelling to guide the refinement of clustering while training.

3. Problem Statement

Existing DEC algorithms enhance learning of representations via structural graph modeling and semantic embedding. Nonetheless, numerous methods are based on fixed neighborhood graphs, and do not explicitly align semantic and structural representations, which are sensitive to graph noise, and constrain flexibility during training. It is necessary to have a common clustering model that dynamically redefines the relationships of documents and imposes semantic-structural consistency to enhance robustness and stability of clustering.

4. Scope of the Study

This study focuses on improving deep embedded document clustering through the integration of semantic representation learning and structural neighborhood modeling. The research scope is defined as follows:

- ❖ **Domain & Problem Focus**
The study is limited to unsupervised document clustering using textual data representations and does not extend to supervised text classification, image clustering, multimodal learning, or other non-textual domains.
- ❖ **Datasets**
Experiments in Thesis-I are conducted on benchmark document clustering datasets including AGNews, BBC News, Reuters, 20-Newsgroups and DBPedia. The study does not investigate domain-specific corpora such as medical, legal, or financial datasets.
- ❖ **Models & Techniques**
Thesis-I focuses on establishing reproducible clustering baselines using KMeans, IDEC, and SDCN with TF-IDF and SBERT-based document representations. Thesis-II extends this framework through dynamic KNN graph reconstruction, adaptive semantic refinement, cross-view contrastive alignment, and joint semantic-structural optimization strategies.
- ❖ **Graph Construction**
The research is limited to cosine similarity-based K-Nearest Neighbor (KNN) graph construction and does not investigate alternative graph generation approaches such as citation networks, knowledge graphs, or entity-relation graph construction.
- ❖ **Evaluation**
Clustering performance is evaluated using Accuracy (ACC), Normalized Mutual Information (NMI), Adjusted Rand Index (ARI), and Cluster Purity. The study focuses on clustering quality evaluation and does not extend to

downstream NLP applications such as information retrieval, question answering, or summarization.

❖ **Exclusions**

The research does not investigate multilingual document clustering, streaming or real-time clustering systems, supervised transformer fine-tuning using labeled data, or large-scale distributed graph processing frameworks.

5. Related Work

Initial advances in document clustering were concerned with the use of the improved text representations. The article "*The performance of BERT as data representation of text clustering*" [9] demonstrated that contextual BERT embeddings had remarkable results in comparison with other feature representations like TF-IDF on various clustering methods. It also pointed out that performance of clustering module is very sensitive to pooling and normalization strategies. This shows that the quality of representation does not ensure stable clustering. In order to mitigate this situation, the frameworks of deep embedded clustering (DEC) were proposed. Structural Deep Clustering Network (SDCN) [1] offered a joint optimization model that combines the autoencoders-based representation learning with clustering by using a student's t-distribution based soft assignments and KL divergence. This represented a shift from two stage clustering to a unified end to end optimization technique. This was later further extended in subsequent work like Deep Structural Enhanced Network for Document Clustering (DSEDC) [2], which added further structural information in the form of graphs to the learning process and enhanced the strength to cluster. Improved Deep Embedded Clustering with Local Structure Preservation (IDEC) [11] prolonged the original DEC idea by jointly optimizing the clustering objective and reconstruction loss of an under-complete autoencoder to preserve local manifold structure. IDEC, in contrast to DEC, integrates a KL-divergence clustering loss with reconstruction constraints, results in improving clustering accuracy and stability across different datasets such as; MNIST, USPS, and REUTERS.

The establishment of structural relations between documents became one of the leading strategies in enhancing the quality of clustering. Graph Convolutional Networks (GCNs) were combined with autoencoders in SDCN [1] to learn deep structural information. Similarly, DSEDC [2] used the semantic representations and structural features learned by GCNs with external, internal semantics, and optimized them together. Attention based Graph Clustering Network (AGCN) [3] added attention to dynamically combine attribute and structural representations, which overcomes over-smoothing in graph learning. These studies indicate that the use of neighborhood information can improve the performance of clustering. In the paper Feature-aware Unsupervised Learning with Joint Variational Attention and Automatic Clustering [13], a variational attention mechanism with joint optimization framework and automatic clustering were proposed. It trained latent feature distributions and attention vectors

together with a KL-based clustering constraint, therefore showing better ACC and NMI compared to DEC, IDEC, and VaDE but with multi-loss objectives. They however also report that they are sensitive to the quality of the constructed graphs and that noise propagation may occur along structural layers. More recently, Graph-Convolutional Networks: Named Entity Recognition and Large Language Model Embedding in Document Clustering [6] gave embedding based on the LLM and entity-based graph building, with better results compared with conventional KNN-based techniques. However, this method involves delicate threshold tuning when constructing graphs, which suggests that structural modeling is hyperparameter sensitive. ARGAn and ARVGA frameworks [14], proposed adversarial regularized graph autoencoders with graph convolutional network encoders. These models use adversarial training methods to impose consistency between latent embeddings and known prior distributions. Empirical findings show that both robustness and clustering performance are improved on citation network data. The method, however, has a high computing complexity during training as well as it is sensitive to the architecture of the decoder and the selection of the prior distribution.

Transformer-based embeddings have become central to document clustering with the success of contextual models as they have effectiveness beyond clustering tasks. A BERT-based framework with attention aggregation and GRU-derived patient embeddings [16] achieved high precision and F1-scores in supervised medical report classification by reinforcing the capacity of contextual transformers to capture long-range semantic dependencies besides being limited to domain-specific supervised settings. BERT embeddings were combined with a DEC-style clustering system with Semantic Deep Embedded Clustering (SDEC) [4], which optimized representation learning along with cluster assignments. This was better at semantic preservation, and was said to be expensive to compute and sensitive to initialization. Further Structural Semantic Network Deep Transformer-based Structural Semantic Network (DTSN) [5] added GCN-based structural modeling to Transformer encoders and combined semantic and structural embeddings in a combined optimization framework. Although it has shown good clustering performance, it has been reported to have weaknesses such as relying on the quality of KNN graph, scaling limitations and assumption of static embedding. A recent hybrid RNN-VAE framework [15] combined sequential dependency modeling using recurrent neural networks and latent feature learning using variational autoencoders for clustering documents. The hybrid architecture worked very well for certain datasets like BBC Sports and Reuters; the ARI and accuracy was high. It however had issues in dealing with highly heterogeneous corpora such as 20-Newsgroups and this indicates that it cannot generalize well with varying semantic distributions. These observations highlight the need for dynamically reconstructed neighborhood graphs that evolve alongside semantic representations during training.

In addition to direct graph modeling, other works enhanced clustering through better embedding geometry. These were complemented with Deep Multi-View Document Clustering with Enhanced Semantic Embedding (MDCE) [7], which enhanced

semantics by relying on neighbor-based and cross-view autoencoders, then clustering with the DEC approach. This method solved the view inconsistency and resulted in the enhancement of convergence stability. Neighborhood Preserving Embedding (NPE) was introduced by the refined SBERT [8] to re-embed sentence representations into a manifold space, followed by the local geometric structure, and then subjecting the inputs to K-means clustering. Though effective to enhance semantic consistency, it needs extra computational procedures and hyperparameter optimization. A generative clustering approach proposed by Variational Deep Embedding (VaDE) [12], integrates a Variational Autoencoder with a Gaussian Mixture prior, modeling the data generation process explicitly and performing clustering through ELBO maximization. This approach outperformed DEC and traditional baselines across multiple datasets and introduced cluster conditional sample generation capability while requiring careful pretraining to avoid poor local minima. The approaches demonstrate that the performance of clustering can be improved by the improved structure and geometry even in the absence of explicit structural graphs.

Recently, contrastive learning has been used to develop better representations. Domain-Adapted Contrastive Learning [10] used contrastive goals together with domain adaptation to enhance low-resource language clustering. Though it greatly enhanced the embedding quality, clustering was carried out in a sequential manner through HAC and K-means and not in an integrated deep clustering approach. This indicates that contrastive learning enhances semantic representation but it is seldom directly incorporated into joint semantic-structural deep clustering systems.

There are a number of recurring trends in the literature:

- ❖ Student's t-distribution and KL divergence to cluster into deterministic coherent clusters are popularly used in the DEC-style [1], [2], [3], [4], [5], [7].
- ❖ GCNs improve the quality of clustering, yet they are vulnerable to the quality of graph construction [1], [5], [6].
- ❖ Transformer-based embeddings are better at modeling semantics over long distances, but they present computational complexity and stability issues [4], [5].
- ❖ Contrastive learning yields better representation matching but is not used with unified deep clustering optimization [10].
- ❖ Multi-view and manifold refinements are better at embedding geometry, but nonetheless do not resolve dynamic structural learning [7], [8].

These results suggest that, although semantic modeling and structural learning have been considerably developed, current options tend to use the neighborhood graph as a static element and insufficient interaction of both semantic and structural perspectives. This encourages more studies on the possibility of coming up with unified models that dynamically model document relationships and still be semantically consistent.

6. Literature Summary Table

Reference	Year	Merits	Demerits	Dataset	Evaluation Metrics
[1]	2020	Integrates AE and GCN in unified deep clustering, dual self-supervised KL optimization and strong gains across 6 datasets.	Sensitive to KNN graph quality, added model complexity and hyperparameter tuning.	USPS HHAR Reuters ACM DBLP Citeseer	ACC NMI ARI F1
[2]	2023	Enhances AE semantics using GCN structural info, layer-wise ensemble reinforcement and strong multi dataset improvement.	Early/late GCN layers may introduce noise and performance degradation on large/dense graphs.	ACM Citeseer Reuters BBC Aminer-s	ACC NMI ARI
[3]	2021	Attention based fusion of attribute and structural features, multiscale representation learning and robust on low quality graphs.	Limited gains on well-connected graphs and middle layers prone to over-smoothing.	USPS HHAR Reuters ACM Citeseer DBLP	ACC NMI ARI F1
[4]	2025	Joint transformer based semantic learning and DEC clustering, semantic refinement improves performance and SOTA on 5 benchmarks.	Requires predefined cluster number, high computational cost and sensitive to weights initialization.	AG News Yahoo! Answers DBPedia Reuters-2 Reuters-5	ACC NMI ARI
[5]	2025	Transformer GCN fusion for inter-document semantics, layer-wise fusion with reinforcement learning and strong results on BC/ACM.	Dependent on KNN graph quality, scalability/generalization issues and static embeddings.	BBC ACM	ACC NMI ARI

[6]	2025	NER based graph construction with LLM embeddings, joint embedding and clustering optimization and outperforms KNN graphs.	Graph sparsity sensitive to thresholding requires careful tuning of τ and p .	BBC News MLSUM Arxiv-10 PubMed	ACC NMI ARI
[7]	2021	Multi-view semantic enhancement via NAE/CAE, DEC-based joint optimization and stable convergence.	Higher computational cost, weaker on small/high-class datasets.	Aminer Aminer (700) 3-sources Reuters Movies multi-source news	NMI ACC
[8]	2023	Manifold refinement of SBERT embeddings, improved semantic consistency and consistent gains across tasks.	Hyperparameter sensitive and added computational cost.	STS12-16 STS-B SICK-R MR CR SUBJ MPQA SST2 TREC MRPC 20 Newsgroups	Spearman ACC NMI ARI
[9]	2022	Systematic BERT vs TF-IDF comparison, strong gains in 28/36 metrics and highlights pooling/normalization impact.	Inconsistent gains on R2 with DEC/IDEC, sensitive to normalization strategy.	AG News (4000) Yahoo! Answers (10000) Reuters R2 (5859)	ACC NMI ARI
[10]	2025	Domain adapted contrastive learning, large scale Greek corpora and approx. 50% clustering gains.	Heavy domain dependence, high computational cost, hyperparameter sensitive and not integrated deep clustering.	Greek internet/social/press corpora (1.59M 230,530 triplets 283,052 docs)	SC Cosine Manhattan Euclidean

[11]	2017	Preserves local structure via reconstruction, joint KL and AE loss and outperforms DEC.	Slower convergence requires γ tuning.	MNIST (70000) USPS (9298) REUTERS-10K (10000)	ACC NMI
[12]	2017	VAE with GMM prior for generative clustering, strong ACC across 5 datasets and cluster conditional generation.	Requires pretraining, risk of poor local minima in early training.	MNIST HHAR REUTERS-10K REUTERS STL-10	ACC KNN err
[13]	2020	Joint variational attention and clustering, multi-loss optimization and superior ACC/NMI over DEC/IDEC/VaDE.	Hyperparameter λ sensitive and complex multi-loss balancing.	Fashion-MNIST CIFAR-10 USPS 20NEWS REUTERS Stack Overflow	ACC NMI
[14]	2019	Adversarial regularization improves embedding robustness with superior link prediction and clustering.	Increased training time, sensitive to decoder/prior choice.	Cora Citeseer PubMed	ACC NMI F1 ARI AUC AP
[15]	2024	Hybrid RNN-VAE captures sequential semantics, strong ARI/ACC on Sports/Reuters and robust to noise.	Poor generalization on heterogeneous datasets (20-Newsgroups).	BBC Sports Reuters-21578 20-Newsgroups	ARI NMI Completeness Homogeneity V-measure ACC
[16]	2023	Effective contextual modeling with BERT+GRU, high P/R/F1 in medical classification.	Data specific, requires larger validation and not clustering.	1000 cancer radiology reports	ACC Precision Recall F1

7. Proposed Methodology

- ❖ **Benchmark Dataset:** Evaluation and comparison will be conducted using datasets in public document clustering (e.g., BBC, Reuters, DBPedia, AG News).
- ❖ **Data Preparation:** Raw documents will be cleaned, and tokenized by Transformer tokenizer before being converted into model-ready inputs.
- ❖ **Semantic Representation Learning:** Transformer encoder will generate contextualized document embeddings representing long-range semantic dependencies that will allow representation learning to adapt to the joint optimization objective that combines clustering and structural objectives.
- ❖ **Dynamic KNN Graph Construction:** A KNN graph based on similarity will be built using semantic embeddings and periodically reconstructed every k epoch(s) during training.
- ❖ **Structural Representation Learning through GCN:** The GCN will learn the global structural relationships among documents by propagating information across dynamically reconstructed graphs.
- ❖ **Cross-View Contrastive Alignment:** A batch contrastive loss will be used to match semantic and structural embeddings in order to impose representation consistency.
- ❖ **Representation Fusion:** The semantic and structural representations will be fused to get unified document representations.
- ❖ **Deep Embedded Clustering:** The student's t-distribution-based soft assignment with the optimization of the KL-divergence will optimize the cluster assignment.
- ❖ **Joint Optimization:** The model will be trained in an end-to-end fashion with a composite clustering, structural and contrastive loss.
- ❖ **Model Evaluation:** The ACC, NMI and ARI will be used to evaluate performance of clustering, while robustness and stability will be assessed through multiple random seed experiments and sensitivity analysis to graph construction parameters.



Figure 1: Flow Diagram of proposed method

8. Tools and Technologies

Technology	Purpose
Python	Primary programming language used for implementation and experimentation
PyTorch	Deep learning framework used for implementing IDEC and SDCN models
Scikit-learn	Used for KMeans clustering PCA TF-IDF vectorization and evaluation metrics
Sentence-Transformers	Used for generating contextual semantic document embeddings from SBERT
FAISS	Used for efficient K-Nearest Neighbor (KNN) graph construction
NumPy	Used for numerical computations and matrix operations
SciPy	Used for sparse matrix processing and graph normalization
Pandas	Used for experiment logging result aggregation and data analysis
Matplotlib	Used for visualization of experimental results and trends
Jupyter Notebook	Used for experimentation iterative development and result analysis
Git & GitHub	Used for version control and research code management

9. Progress & Initial Results

The initial phase of the research focused on implementing and evaluating baseline document clustering systems using both traditional and deep clustering approaches.

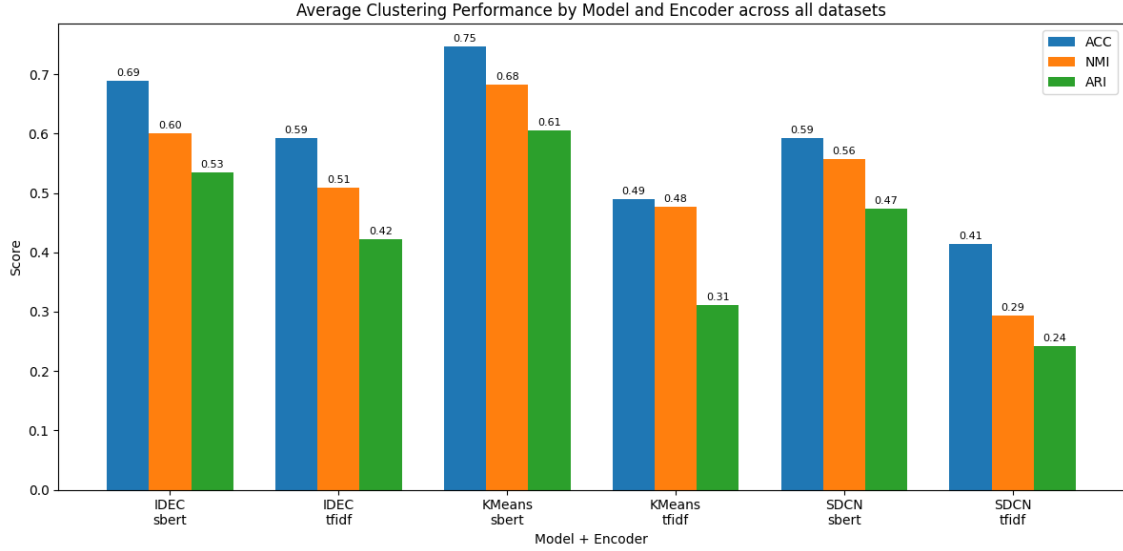
The following components have been successfully implemented:

- ❖ TF-IDF embedding pipeline
- ❖ SBERT embedding pipeline
- ❖ KMeans clustering baseline
- ❖ IDEC clustering framework
- ❖ SDCN clustering framework
- ❖ KNN graph construction using cosine similarity
- ❖ Graph normalization and sparse adjacency processing
- ❖ Automated clustering evaluation framework
- ❖ Experiment logging and metric tracking pipeline

The experimentation framework supports reproducible evaluation across multiple datasets and embedding configurations.

Encoder	ACC	NMI	ARI	Purity
SBERT	0.6763	0.6135	0.538	0.7482
TF-IDF	0.4989	0.4265	0.3253	0.5797

Model	ACC	NMI	ARI	Purity
IDEC	0.6407	0.5545	0.4783	0.7277
KMeans	0.6184	0.5799	0.4587	0.6955
SDCN	0.5036	0.4256	0.3578	0.5686



10. Preliminary Experimental Findings

- ❖ The experimental results indicate that contextual semantic embeddings generated using SBERT consistently outperform traditional TF-IDF representations across most clustering configurations and evaluation metrics.
- ❖ KMeans operating directly on SBERT embeddings achieved highly competitive performance, suggesting that pretrained transformer-based semantic representations already preserve strong latent document separability.
- ❖ Deep clustering architectures such as IDEC and SDCN did not consistently outperform simpler clustering approaches when applied to dense contextual embeddings, indicating possible semantic distortion introduced during latent compression and graph propagation.
- ❖ The performance of graph-based clustering methods was found to be highly sensitive to graph construction quality, embedding characteristics, and neighborhood structure, particularly on large and heterogeneous datasets.
- ❖ The observed limitations of static graph construction and weak semantic-structural alignment motivate the need for adaptive semantic refinement, dynamic graph reconstruction, and contrastive semantic-structural learning mechanisms in Thesis-II.

11. Current Challenges and Limitations

- ❖ **Static Graph Dependency:** Existing graph-based clustering methods rely on fixed graphs that may become semantically outdated as latent representations evolve during training.
- ❖ **Graph Sensitivity:** Clustering performance was found to be highly sensitive to KNN configuration, similarity metrics, graph sparsity, and embedding normalization.
- ❖ **Scalability Constraints:** Large-scale graph construction and propagation introduce significant computational and memory overhead.
- ❖ **Semantic-Structural Misalignment:** Current models jointly optimize semantic and structural representations without explicit mechanisms for semantic-structural alignment.
- ❖ **Training Stability:** Multi-objective deep clustering optimization remains sensitive to initialization quality, graph construction and centroid initialization.

12. MS Thesis-II Goals

The second phase of the research will focus on extending the current baseline framework through adaptive semantic and structural learning mechanisms for deep document clustering.

The primary goals of Thesis Part-II are as follows:

- ❖ **Dynamic Graph Reconstruction**
Develop mechanisms for periodically reconstructing K-Nearest Neighbor KNN graphs during training to better reflect evolving semantic representations.
- ❖ **Transformer-Guided Semantic Representation Refinement**
Investigate semantic representation refinement strategies using transformer-generated contextual embeddings during clustering optimization.
- ❖ **Cross-View Contrastive Alignment**
Introduce contrastive learning objectives to align:
 - a. Semantic embeddings
 - b. Graph-based structural embeddings

❖ **Joint Semantic-Structural Optimization**

Develop a unified optimization framework integrating:

- a. Clustering refinement
- b. Semantic representation learning
- c. Structural consistency learning
- d. Contrastive alignment objectives

❖ **Ablation and Robustness Analysis**

Conduct controlled ablation experiments to evaluate:

- a. Static versus dynamic graph construction
- b. Contrastive versus non-contrastive optimization
- c. Semantic-only versus semantic-structural learning.

❖ **Scalability and Stability Evaluation**

Evaluate convergence stability, robustness, and scalability across all selected benchmark document clustering datasets under varying graph and embedding configurations.

The overall objective of Thesis Part-II is to investigate whether adaptive interaction between semantic representation learning and structural neighborhood modeling can improve clustering robustness and semantic consistency in deep document clustering systems.

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